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School of Computing

Department of Computer Science and Engineering

10211CS224 / FULL STACK APPLICATION DEVELOPMENT (JAVA)

PROJECT REVIEW : MILESTONE PROJECT – 1

Date: 19-04-2026

JOB PORTAL MANAGEMENT SYSTEM

SDG: SDG 9 – Industry, Innovation and Infrastructure

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INTRODUCTION



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- Job Portal Management System is a web-based platform that connects employers and job seekers (students/users) in a centralized environment.
- It simplifies the recruitment process by allowing employers to post job vacancies and manage applications efficiently.
- Job seekers can browse available jobs, upload or update resumes, and apply for suitable positions
- The system provides features like shortlisting candidates, making hiring faster and more organized.
- It reduces the need for manual recruitment processes and improves communication between employers and applicants.
- The platform ensures easy access to job opportunities anytime and anywhere.
- It enhances transparency, reduces hiring time, and improves overall efficiency in the recruitment process.
- This system supports employment growth and aligns with SDG 8: Decent Work and Economic Growth by promoting job accessibility.

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PROBLEM STATEMENT



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In today's competitive job market, students and job seekers often struggle to find relevant job opportunities in a timely manner, while employers face difficulty in identifying suitable candidates efficiently. The existing recruitment processes are often fragmented, time-consuming, and lack a centralized platform for seamless interaction between job providers and applicants.

To address these challenges, a Job Portal Management System is required to provide an

integrated solution where employers can host job openings and shortlist candidates, and students/job seekers can browse jobs, upload or update resumes, and apply for suitable positions. The system aims to streamline the recruitment process by enabling easy communication, efficient candidate management, and quick access to opportunities.

OBJECTIVE(S)

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- **Centralized Web Platform:** A single, unified hub connecting job seekers and employers directly.
- **Role-Based Access Control:** Dedicated, secure dashboards for Students, Employers, and Admins.
- **Modern & Responsive UI:** Built with React.js using an engaging, high-contrast Neo-Brutalist design.
- **Automated Tracking:** Replaces messy email applications with real- time status updates (Pending, Shortlisted, Rejected).
- **High Security:** Implements Spring Security & JWT for stateless, secure authentication and data protection.
- **Scalable Architecture:** RESTful APIs using Spring Boot and MySQL allow for fast data retrieval and future expansion.
- **Streamlined Hiring Workflow:** Enables digital resume uploads, advanced job filtering, and instant applicant reviews.

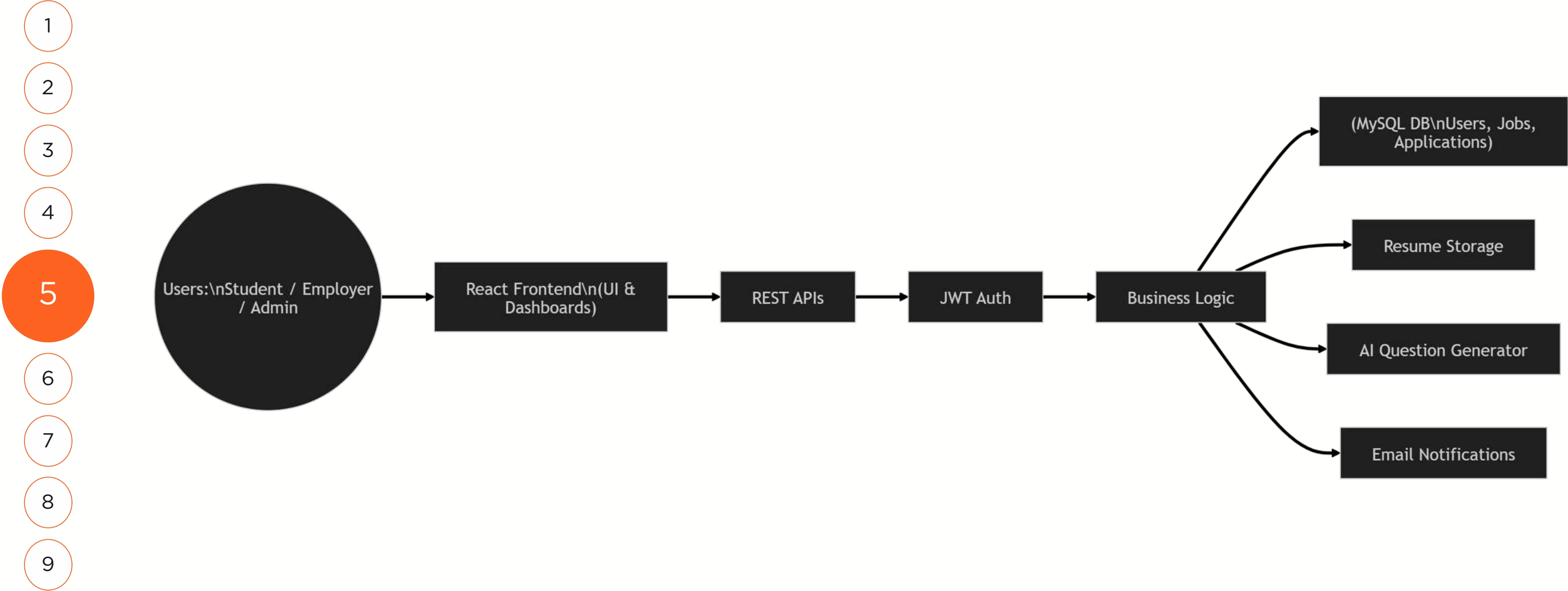
PROPOSED SYSTEM



- **Full-Stack Architecture** : A web application using Spring Boot and React to connect employers and job seekers efficiently.
- **Unified Recruitment Workflow** : Combines job posting, applicant tracking, shortlisting, and resume management in one system.
- **AI-Powered Employer Tools** : Provides an AI-based question generator for creating role-specific technical assessments.
- **Eliminates Hiring Fragmentation** : Centralizes recruitment processes to reduce scattered workflows and improve organization.
- **Resolves Communication Delays** : Offers real-time updates and automated notifications for better user communication.
- **Superior Modern UI/UX** : Uses a responsive Neo-Brutalist design for a clean and engaging user experience. **Distinct Role-Based Dashboards** : Provides separate dashboards for students and employers for better usability. **Secure and Scalable Ecosystem** : Uses JWT authentication and admin controls to ensure security and scalability.

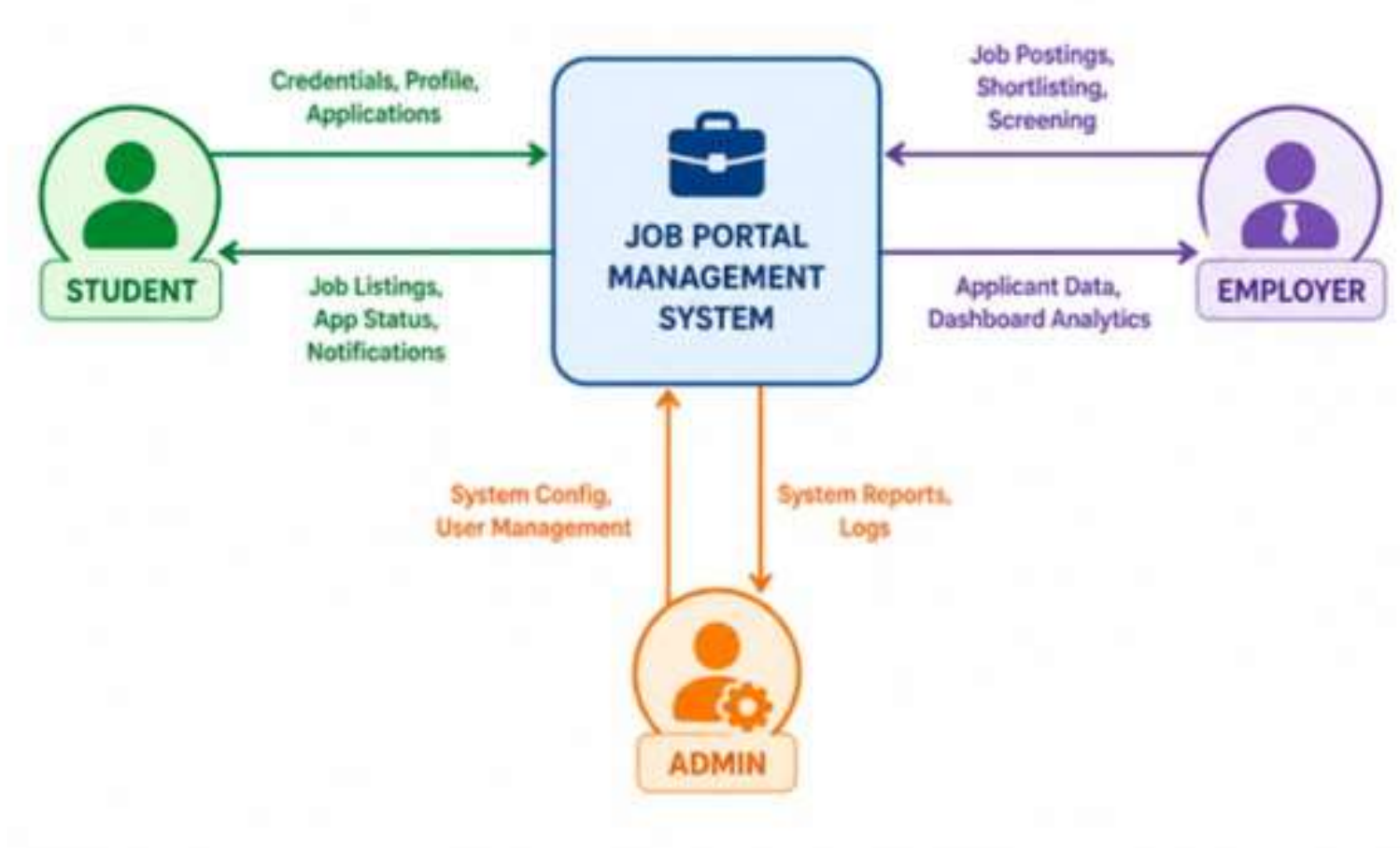
PROPOSED SYSTEM ARCHITECTURE

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WORKFLOW

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TECHNOLOGIES USED



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Frontend : Built using React.js with Vite for fast development and optimized performance. React Router enables seamless role-based navigation, while Tailwind/Vanilla CSS provides a modern Neo- Brutalist UI.

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Axios is used for efficient communication with backend APIs.

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Backend : Developed using Java and Spring Boot to create scalable and secure REST APIs. Spring Security handles JWT-based authentication and role authorization. JPA/Hibernate simplifies database operations and ensures efficient data persistence.

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Database : MySQL is used as a relational database to store user details, job postings, and application data. It ensures data consistency, structured storage, and efficient querying. The system supports reliable data management for recruitment workflows.

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Third-Party APIs : Google Gemini API is integrated to generate dynamic, role-specific interview questions. SMTP services using Gmail are used for sending automated emails and OTP verification. This improves communication and user engagement.

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Tools : Development is done using Visual Studio Code and IntelliJ IDEA for coding and debugging. Git and GitHub manage version control. Postman and Swagger UI are used for API testing and documentation.

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Package Managers : npm is used to manage frontend dependencies and streamline development. Maven handles backend build processes, dependency management, and project configuration. Together, they ensure efficient and organized development workflows.

MODULE DESCRIPTION



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Module 1: User Authentication:

- Secure login and registration
- Handles secure login by verifying username and password
- Role-based access (Student / Employer / Admin)

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Module 2: Job Management:

- Employers can post, edit, delete jobs

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Module 3: Application Management:

- Students apply for jobs, Track application status

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Module 4: Search & Filter:

Filter jobs by:

- Location
- Skills
- Experience

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IMPLEMENTATION SCREENSHOTS

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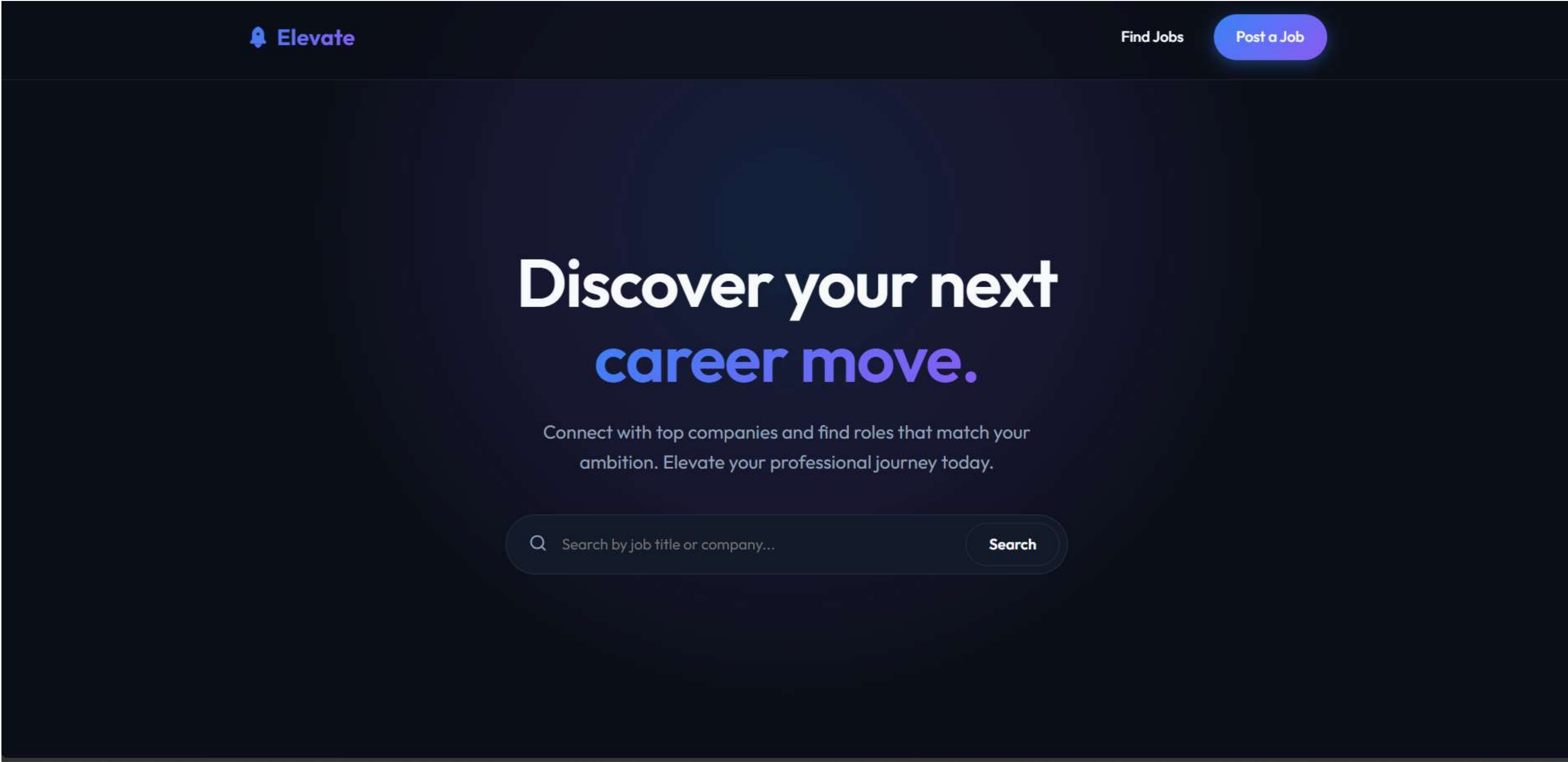
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IMPLEMENTATION SCREENSHOTS



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Elevate

Find Jobs

Post a Job

Hiring? Post a Job

Reach thousands of professionals actively looking for their next role.

Job Title

e.g. Senior Product Designer

Company Name

e.g. TechNova

Location

e.g. San Francisco (Remote)

Salary Range (Optional)

e.g. \$120k - \$150k

Job Description

Describe the role and responsibilities...

Publish Listing →

IMPLEMENTATION SCREENSHOTS

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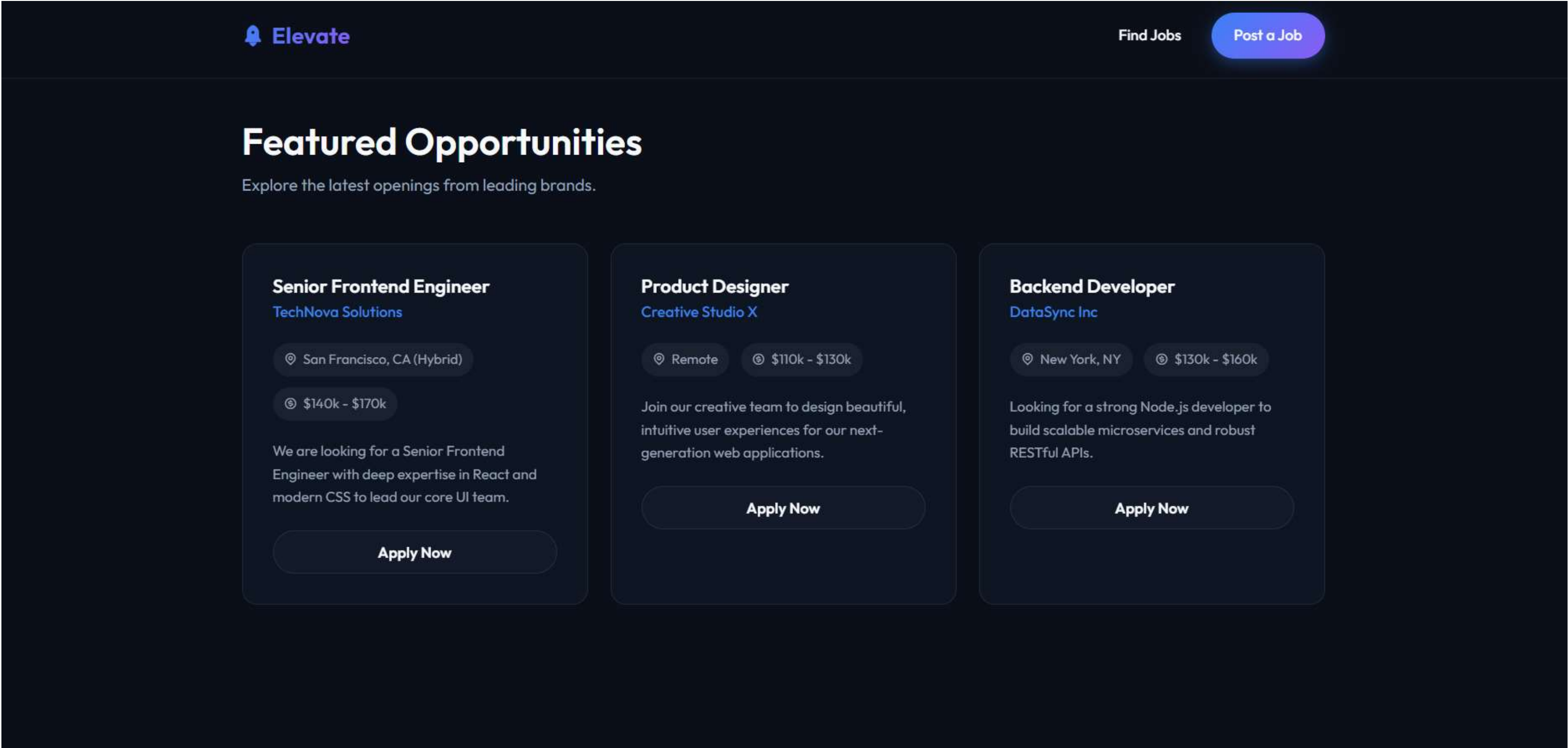
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ADVANTAGES & LIMITATIONS



ADVANTAGES:

- User-friendly interface
- Fast and efficient system
- Centralized data management
- Secure authentication

LIMITATIONS:

- Requires internet connection
- Basic features in current version
- No AI-based recommendations
- Limited scalability without cloud

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Thank you